DR. MAHALINGAM COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Career Planning and Guidance

Aptitude & Logical Reasoning Assessment (APT60)

OFFICIAL SOLUTION KEY & EXPLANATIONS

OFFICIAL ANSWER KEY

Assessment Code: APT60
Duration: 60 Minutes
Total Questions: 60 MCQs
Marking: +1.0 Mark | -1.0 Negative Mark

QUICK REFERENCE ANSWER KEY (QUESTIONS 1 – 60)

Q1 **A**	Q2 **D**	Q3 **C**	Q4 **D**	Q5 **D**	Q6 **B**	Q7 **B**	Q8 **D**	Q9 **C**	Q10 **D**
Q11 **C**	Q12 **D**	Q13 **B**	Q14 **B**	Q15 **B**	Q16 **B**	Q17 **B**	Q18 **C**	Q19 **A**	Q20 **B**
Q21 **C**	Q22 **B**	Q23 **C**	Q24 **C**	Q25 **C**	Q26 **B**	Q27 **B**	Q28 **B**	Q29 **B**	Q30 **A**
Q31 **A**	Q32 **B**	Q33 **B**	Q34 **C**	Q35 **B**	Q36 **D**	Q37 **A**	Q38 **C**	Q39 **B**	Q40 **D**
Q41 **D**	Q42 **C**	Q43 **B**	Q44 **C**	Q45 **D**	Q46 **B**	Q47 **B**	Q48 **C**	Q49 **C**	Q50 **C**
Q51 **D**	Q52 **B**	Q53 **B**	Q54 **B**	Q55 **B**	Q56 **A**	Q57 **A**	Q58 **B**	Q59 **A**	Q60 **B**

Question 1 [+1 Mark, -1 Negative Mark]

Three numbers are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:

- **A** 40 ✓ Correct Answer
- **B** 80
- **C** 120
- **D** 200

Step-by-Step Solution: Let the numbers be 3x, 4x, and 5x. Their LCM is 60x = 2400 => x = 40. The HCF of 3x, 4x, 5x is x, which is 40.

Question 2 [+1 Mark, -1 Negative Mark]

The ratio of two numbers is 3 : 4 and their H.C.F. is 4. Their L.C.M. is:

- **A** 12
- **B** 16
- **C** 24
- **D** 48 ✓ Correct Answer

Step-by-Step Solution: The numbers are 3 * 4 = 12 and 4 * 4 = 16. LCM(12, 16) = 48.

Question 3 [+1 Mark, -1 Negative Mark]

The L.C.M. of two numbers is 48. The numbers are in the ratio 2 : 3. Then the sum of the numbers is:

- **A** 28
- **B** 32
- **C** 40 ✓ Correct Answer
- **D** 64

Step-by-Step Solution: Let the numbers be 2x and 3x. Their LCM is 6x = 48 => x = 8. The numbers are 16 and 24. Sum = 16 + 24 = 40.

Question 4 [+1 Mark, -1 Negative Mark]

If one-third of one-fourth of a number is 15, then three-tenth of that number is:

A	35	B	36
C	45	D	54 ✓ Correct Answer

Step-by-Step Solution: (1/3) * (1/4) * N = 15 => N / 12 = 15 => N = 180. Therefore, (3/10) * 180 = 54.

Question 5 [+1 Mark, -1 Negative Mark]

Three times the first of three consecutive odd integers is 3 more than twice the third. The third integer is:

A	9	B	11
C	13	D	15 ✓ Correct Answer

Step-by-Step Solution: Let the integers be x, x + 2, x + 4. 3x = 2(x + 4) + 3 => 3x = 2x + 11 => x = 11. The third integer is 11 + 4 = 15.

Question 6 [+1 Mark, -1 Negative Mark]

The difference between a two-digit number and the number obtained by interchanging the positions of its digits is 36. What is the difference between the two digits of that number?

A	3	B	4 ✓ Correct Answer
C	9	D	Cannot be determined

Step-by-Step Solution: Difference = (10x + y) - (10y + x) = 9(x - y) = 36 => x - y = 4.

Question 7 [+1 Mark, -1 Negative Mark]

A two-digit number is such that the product of the digits is 8. When 18 is added to the number, then the digits are reversed. The number is:

A	18	B	24 ✓ Correct Answer
C	42	D	81

Step-by-Step Solution: Let the number be 10x + y. xy = 8. Adding 18 reverses digits: 10x + y + 18 = 10y + x => 9(y - x) = 18 => y - x = 2. With xy = 8 and y - x = 2, we get x = 2, y = 4. The number is 24.

Question 8 [+1 Mark, -1 Negative Mark]

If the number 481 * 673 is completely divisible by 9, then the smallest whole number in place of * will be:

A	2	B	5
C	6	D	7 ✓ Correct Answer

Step-by-Step Solution: Sum of digits = 4 + 8 + 1 + * + 6 + 7 + 3 = 29 + *. The nearest multiple of 9 greater than or equal to 29 is 36. Thus, * = 36 - 29 = 7.

Question 9 [+1 Mark, -1 Negative Mark]

The value of a washing machine depreciates at the rate of 10% every year. If its present value is Rs. 8,748, then what was the price of the washing machine three years ago?

A Rs. 10,000

B Rs. 11,000

C Rs. 12,000 ✓ Correct Answer

D Rs. 15,000

Step-by-Step Solution: P * (1 - 0.10)^3 = 8748 => P * 0.729 = 8748 => P = 8748 / 0.729 = Rs. 12,000.

Question 10 [+1 Mark, -1 Negative Mark]

A student erroneously multiplied a number by 2/5 instead of 5/2. What is the percentage error in the calculation?

A 24%

B 54%

C 74%

D 84% ✓ Correct Answer

Step-by-Step Solution: Error = (5/2 - 2/5) / (5/2) * 100 = (21/10) * (2/5) * 100 = (21/25) * 100 = 84%.

Question 11 [+1 Mark, -1 Negative Mark]

Ram spends 70% of his salary and deposits 15% of his salary in the bank. If he is left with Rs. 1,500, what is his monthly salary?

A Rs. 9,000

B Rs. 9,500

C Rs. 10,000 ✓ Correct Answer

D Rs. 10,500

Step-by-Step Solution: Remaining percentage = 100% - (70% + 15%) = 15%. 15% of salary = Rs. 1,500 => Salary = 1500 / 0.15 = Rs. 10,000.

Question 12 [+1 Mark, -1 Negative Mark]

A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had:

A 588 apples

B 600 apples

C 672 apples

D 700 apples ✓ Correct Answer

Step-by-Step Solution: Remaining apples = 100% - 40% = 60%. 60% of Total = 420 => Total = 420 / 0.60 = 700 apples.

Question 13 [+1 Mark, -1 Negative Mark]

The price of an article was first decreased by 20% and then increased by 15%. What was the overall percent decrease or increase?

A 8% increase

B 8% decrease ✓ Correct Answer

C 2% decrease

D 2% increase

Step-by-Step Solution: Net factor = (1 - 0.20) * (1 + 0.15) = 0.80 * 1.15 = 0.92 => 1 - 0.92 = 0.08 = 8% decrease.

Question 14 [+1 Mark, -1 Negative Mark]

A train travelled a certain distance at a uniform speed. Had the speed been 6 km/h more, it would have needed 4 hours less. Had the speed been 6 km/h less, it would have needed 6 hours more. The distance, in km, travelled by the train is:

A 620 km

B 720 km ✓ Correct Answer

C 540 km

D 840 km

Step-by-Step Solution: Using D = s*(s+6)*4/6 = s*(s-6)*6/6 => 4(s+6) = 6(s-6) => 4s + 24 = 6s - 36 => 2s = 60 => s = 30 km/h. Distance D = 30 * 36 * 4 / 6 = 720 km.

Question 15 [+1 Mark, -1 Negative Mark]

A boat's speed in still water is 22 km/h, while the river is flowing with a speed of 8 km/h. The time taken to cover a certain distance upstream is 4 hours more than the time taken to cover the same distance downstream. Find the distance.

A 100 km

B 105 km ✓ Correct Answer

C 110 km

D 115 km

Step-by-Step Solution: Downstream speed = 22 + 8 = 30 km/h. Upstream speed = 22 - 8 = 14 km/h. D/14 - D/30 = 4 => D * (16 / 420) = 4 => D = 4 * 420 / 16 = 105 km.

Question 16 [+1 Mark, -1 Negative Mark]

A boat goes 20 km upstream and 44 km downstream in 8 hours. In 5 hours, it goes 15 km upstream and 22 km downstream. Determine the speed of the boat in still water.

A 5 km/h

B 8 km/h ✓ Correct Answer

C 10 km/h

D 12 km/h

Step-by-Step Solution: Let 1/u = x and 1/v = y. 20x + 44y = 8 and 15x + 22y = 5 => 30x + 44y = 10. Subtracting gives 10x = 2 => x = 1/5 (upstream = 5 km/h) and y = 1/11 (downstream = 11 km/h). Speed in still water = (5 + 11)/2 = 8 km/h.

Question 17 [+1 Mark, -1 Negative Mark]

A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is:

A 5/7

B 8/15 ✓ Correct Answer

C 9/16

D 3/10

Step-by-Step Solution: A's 1-day work = 1/15, B's 1-day work = 1/20. Together in 1 day = 1/15 + 1/20 = 7/60. In 4 days = 4 * 7/60 = 7/15. Work left = 1 - 7/15 = 8/15.

Question 18 [+1 Mark, -1 Negative Mark]

A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

A 5 days

B 10 days

C 15 days ✓ Correct Answer

D 20 days

Step-by-Step Solution: Day 1 (A) = 1/20. Day 2 (A) = 1/20. Day 3 (A+B+C) = 1/20 + 1/30 + 1/60 = 6/60 = 1/10. Work in 3 days = 1/20 + 1/20 + 1/10 = 1/5. Entire work completes in 5 * 3 = 15 days.

Question 19 [+1 Mark, -1 Negative Mark]

A:B is 1:2; B:C is 3:2 and C:D is 1:3. Find A:B:C:D.

A 3:6:4:12 ✓ Correct Answer

B 2:3:5:7

C 3:5:7:6

D 2:1:6:13

Step-by-Step Solution: A:B = 1:2 = 3:6. B:C = 3:2 = 6:4. Thus A:B:C = 3:6:4. C:D = 1:3 = 4:12. Thus A:B:C:D = 3:6:4:12.

Question 20 [+1 Mark, -1 Negative Mark]

The ratio of two numbers is 11 : 12. If the sum of the numbers is 253, find the difference between the numbers.

A 10

B 11 ✓ Correct Answer

C 12

D 21

Step-by-Step Solution: 11x + 12x = 23x = 253 => x = 11. Difference = 12x - 11x = x = 11.

Question 21 [+1 Mark, -1 Negative Mark]

A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1,000 more than D, what is B's share?

A Rs. 500

B Rs. 1,500

C Rs. 2,000 ✓ Correct Answer

D None of these

Step-by-Step Solution: C's share - D's share = 4x - 3x = x = Rs. 1,000. B's share = 2x = 2 * 1,000 = Rs. 2,000.

Question 22 [+1 Mark, -1 Negative Mark]

The average weight of A, B and C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, then the weight of B is:

A 30 kg

B 31 kg ✓ Correct Answer

C 32 kg

D 33 kg

Step-by-Step Solution: A + B + C = 45 * 3 = 135 kg. A + B = 80 kg => C = 135 - 80 = 55 kg. B + C = 86 kg => B = 86 - 55 = 31 kg.

Question 23 [+1 Mark, -1 Negative Mark]

The average weight of 36 men is 50 kg. It was found later that the figure of 37 kg was misread as 73 kg. What is the correct average?

A 40 kg

B 45 kg

C 49 kg ✓ Correct Answer

D 50 kg

Step-by-Step Solution: Total weight miscalculated by (73 - 37) = +36 kg. Correct average = 50 - (36 / 36) = 50 - 1 = 49 kg.

Question 24 [+1 Mark, -1 Negative Mark]

The average age of 30 students is 9 years. If the teacher is included the average becomes 10 years. What is the age of the teacher?

A 30 years

B 35 years

C 40 years ✓ Correct Answer

D 45 years

Step-by-Step Solution: Total age of 30 students = 270. Total age with teacher (31 people) = 31 * 10 = 310. Teacher's age = 310 - 270 = 40 years.

Question 25 [+1 Mark, -1 Negative Mark]

Five years ago, the ages of A and B were 1 : 2. At present the ratio is 3 : 4. Find the present age of A.

A 5 years

B 6 years

C 7.5 years ✓ Correct Answer

D 8 years

Step-by-Step Solution: Let present ages be 3x and 4x. (3x - 5) / (4x - 5) = 1/2 => 6x - 10 = 4x - 5 => 2x = 5 => x = 2.5. A's present age = 3 * 2.5 = 7.5 years.

Question 26 [+1 Mark, -1 Negative Mark]

Present ages of Sam and Anand are in the ratio of 5 : 4 respectively. Three years hence, the ratio of their ages will become 11 : 9 respectively. What is Anand's present age in years?

A 15 years

B 24 years ✓ Correct Answer

C 30 years

D 32 years

Step-by-Step Solution: (5x + 3) / (4x + 3) = 11/9 => 45x + 27 = 44x + 33 => x = 6. Anand's present age = 4x = 24 years.

Question 27 [+1 Mark, -1 Negative Mark]

A father said to his son, 'I was as old as you are at the present at the time of your birth'. If the father's age is 38 years now, the son's age five years back was:

A 12 years

B 14 years ✓ Correct Answer

C 15 years

D 20 years

Step-by-Step Solution: Father's age at son's birth = 38 - x = x => 2x = 38 => x = 19 years. Five years back, son's age = 19 - 5 = 14 years.

Question 28 [+1 Mark, -1 Negative Mark]

A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1,600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:

A Rs. 80

B Rs. 121 ✓ Correct Answer

C Rs. 145

D Rs. 189

Step-by-Step Solution: Half-yearly rate = 2.5%. Amount = 1600 * (1 + 0.025)^2 + 1600 * (1 + 0.025) = 1681 + 1640 = 3321. Total deposited = 3200. Interest = 3321 - 3200 = Rs. 121.

Question 29 [+1 Mark, -1 Negative Mark]

The difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 4% per annum is Re. 1. The sum (in Rs.) is:

A Rs. 597

B Rs. 625 ✓ Correct Answer

C Rs. 720

D Rs. 856

Step-by-Step Solution: Difference = P * (R/100)^2 => 1 = P * (4/100)^2 = P * (1/625) => P = Rs. 625.

Question 30 [+1 Mark, -1 Negative Mark]

A card is drawn from a pack of 52 cards. The probability of getting a queen of club or a king of heart is:

A 1/26 ✓ Correct Answer

B 2/35

C 2/67

D 3/65

Step-by-Step Solution: Total cards = 52. Queen of club = 1 card, King of heart = 1 card. Total favorable outcomes = 2. Probability = 2/52 = 1/26.

Question 31 [+1 Mark, -1 Negative Mark]

From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

A 756 ✓ Correct Answer

B 748

C 789

D 800

Step-by-Step Solution: Ways = (7C3 * 6C2) + (7C4 * 6C1) + (7C5 * 6C0) = (35 * 15) + (35 * 6) + (21 * 1) = 525 + 210 + 21 = 756 ways.

Question 32 [+1 Mark, -1 Negative Mark]

In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

A 625

B 720 ✓ Correct Answer

C 840

D 960

Step-by-Step Solution: Vowels: E, A, I (3 vowels). Consonants: L, D, N, G (4 consonants). Treating vowels as 1 single block yields 5 units, arranged in 5! = 120 ways. The 3 vowels arrange internally in 3! = 6 ways. Total = 120 * 6 = 720.

Question 33 [+1 Mark, -1 Negative Mark]

Rajeev's age after 15 years will be 5 times his age 5 years back. What is the present age of Rajeev?

A 5 years

B 10 years ✓ Correct Answer

C 15 years

D 20 years

Step-by-Step Solution: R + 15 = 5(R - 5) => R + 15 = 5R - 25 => 4R = 40 => R = 10 years.

Question 34 [+1 Mark, -1 Negative Mark]

The ratio of father's age to son's age is 4 : 1. The product of their ages is 196. What will be the ratio of their ages after 5 years?

A 5:11

B 11:6

C 11:4 ✓ Correct Answer

D 11:7

Step-by-Step Solution: (4x) * (x) = 196 => 4x^2 = 196 => x^2 = 49 => x = 7. Father = 28, Son = 7. After 5 years: Father = 33, Son = 12. Ratio = 33:12 = 11:4.

Question 35 [+1 Mark, -1 Negative Mark]

A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:

A Rs. 1,425

B Rs. 1,500 ✓ Correct Answer

C Rs. 1,537.50

D Rs. 1,576

Step-by-Step Solution: Let total profit be P. Remaining after charity = 0.95P. A's share = (3/5) * 0.95P = 0.57P = 855 => P = 855 / 0.57 = Rs. 1,500.

Question 36 [+1 Mark, -1 Negative Mark]

A starts business with Rs. 3,500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?

A Rs. 7,500

B Rs. 8,000

C Rs. 8,500

D Rs. 9,000 ✓ Correct Answer

Step-by-Step Solution: Ratio of investments = (3500 * 12) / (B * 7) = 42000 / 7B = 6000 / B. Given 6000 / B = 2/3 => 2B = 18000 => B = Rs. 9,000.

Question 37 [+1 Mark, -1 Negative Mark]

In what ratio should two varieties of sugar of Rs. 18 per kg and Rs. 24 per kg be mixed together to get a mixture whose cost is Rs. 20 per kg?

A 2 : 1 ✓ Correct Answer

B 3 : 1

C 4 : 1

D 5 : 1

Step-by-Step Solution: By rule of alligation: (24 - 20) : (20 - 18) = 4 : 2 = 2 : 1.

Question 38 [+1 Mark, -1 Negative Mark]

Two vessels A and B contain spirit and water in the ratio 5 : 2 and 7 : 6 respectively. Find the ratio in which this mixture be mixed to obtain a new mixture in the vessel C containing spirit and water in the ratio 8 : 5?

A 5 : 7

B 6 : 7

C 7 : 9 ✓ Correct Answer

D 7 : 3

Step-by-Step Solution: Fraction of spirit: A = 5/7, B = 7/13, Mean C = 8/13. Ratio = (8/13 - 7/13) : (5/7 - 8/13) = (1/13) : (9/91) = (7/91) : (9/91) = 7 : 9.

Question 39 [+1 Mark, -1 Negative Mark]

A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is:

A 10%

B 20% ✓ Correct Answer

C 30%

D 40%

Step-by-Step Solution: Let CP of 1 litre milk be Rs. 1. SP of 1 litre mixture = Rs. 1. Gain = 25% => CP of mixture = 1 / 1.25 = Rs. 0.80. Quantity of milk = 0.80 L, water = 0.20 L. Water percentage = (0.20 / 1.00) * 100 = 20%.

Question 40 [+1 Mark, -1 Negative Mark]

In a 729 litre mixture of milk and water, the ratio of milk to water is 7 : 2. To get a new mixture containing milk and water in the ratio 7 : 3, the amount of water to be added is:

A 50 L

B 65 L

C 80 L

D 81 L ✓ Correct Answer

Step-by-Step Solution: Milk = (7/9) * 729 = 567 L. Water = (2/9) * 729 = 162 L. In new ratio 7:3 with 567 L milk, required water = (3/7) * 567 = 243 L. Water to add = 243 - 162 = 81 L.

Question 41 [+1 Mark, -1 Negative Mark]

At what rate percent per annum will a sum of money double in 20 years?

A 1.2%

B 2%

C 4%

D 5% ✓ Correct Answer

Step-by-Step Solution: SI = P. Rate R = (SI * 100) / (P * T) = (P * 100) / (P * 20) = 5% per annum.

Question 42 [+1 Mark, -1 Negative Mark]

A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:

A Rs. 650

B Rs. 690

C Rs. 698 ✓ Correct Answer

D Rs. 700

Step-by-Step Solution: SI for 1 year = 854 - 815 = Rs. 39. SI for 3 years = 39 * 3 = Rs. 117. Principal = 815 - 117 = Rs. 698.

Question 43 [+1 Mark, -1 Negative Mark]

A shopkeeper sells an article for Rs. 200 with a loss of 20%. Find the cost price of the article.

A Rs. 220

B Rs. 250 ✓ Correct Answer

C Rs. 280

D Rs. 260

Step-by-Step Solution: SP = 0.80 * CP => CP = 200 / 0.80 = Rs. 250.

Question 44 [+1 Mark, -1 Negative Mark]

S.P. of 10 candles is the same as C.P. of 12 candles. Find the gain percent.

A 11%

B 15%

C 20% ✓ Correct Answer

D 25%

Step-by-Step Solution: Gain% = (12 - 10) / 10 * 100 = (2/10) * 100 = 20%.

Question 45 [+1 Mark, -1 Negative Mark]

If a retailer marks up his price by 30% and then allows a 30% discount, what is his percentage profit or loss?

A 9% gain

B 8% gain

C 8% loss

D 9% loss ✓ Correct Answer

Step-by-Step Solution: Net change = 30 - 30 - (30 * 30)/100 = -9% => 9% loss.

Question 46 [+1 Mark, -1 Negative Mark]

Look at this series: 2, 6, 18, 54, ... What number should come next?

A 98

B 162 ✓ Correct Answer

C 195

D 324

Step-by-Step Solution: Each number is multiplied by 3: 2 * 3 = 6, 6 * 3 = 18, 18 * 3 = 54, 54 * 3 = 162.

Question 47 [+1 Mark, -1 Negative Mark]

Look at this series: 22, 21, 23, 22, 24, 23, ... What number should come next?

A 20

B 25 ✓ Correct Answer

C 30

D 45

Step-by-Step Solution: The pattern alternates between -1 and +2: 22 - 1 = 21, 21 + 2 = 23, 23 - 1 = 22, 22 + 2 = 24, 24 - 1 = 23, 23 + 2 = 25.

Question 48 [+1 Mark, -1 Negative Mark]

Find the next number in the series: 3, 8, 18, 38, 78, ?

A 148

B 156

C 158 ✓ Correct Answer

D 160

Step-by-Step Solution: Differences are: 5, 10, 20, 40, ... Next difference is 80 => 78 + 80 = 158.

Question 49 [+1 Mark, -1 Negative Mark]

One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?

A East

B West

C North ✓ Correct Answer

D South

Step-by-Step Solution: In the morning, the sun rises in the East and shadows fall toward the West. Vishal's shadow falls to the West. Since this shadow is to the left of Udai, Udai's left points West, meaning Udai is facing North.

Question 50 [+1 Mark, -1 Negative Mark]

From his house, Lokesh went 15 km to the North. Then he turned West and covered 10 km. Then he turned South and covered 5 km. Finally turning to the East, he covered 10 km. In which direction is he from his house?

A East

B West

C North ✓ Correct Answer

D South

Step-by-Step Solution: East and West movements cancel out (10 km West then 10 km East). Net North-South displacement = 15 km North - 5 km South = 10 km North. He is North of his house.

Question 51 [+1 Mark, -1 Negative Mark]

Rasik walked 20 m towards North. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally he turns left and walks 15 m. In which direction and how many metres is he from the starting position?

A 15 m West

B 30 m East

C 30 m West

D 45 m East ✓ Correct Answer

Step-by-Step Solution: North-South: +20 - 35 + 15 = 0 m. East-West: +30 + 15 = 45 m East. He is 45 m East of the starting position.

Question 52 [+1 Mark, -1 Negative Mark]

Pointing to a photograph of a boy Suresh said, 'He is the son of the only son of my mother.' How is Suresh related to that boy?

A Brother

B Father ✓ Correct Answer

C Uncle

D Cousin

Step-by-Step Solution: The only son of Suresh's mother is Suresh himself. The boy is the son of Suresh, so Suresh is the father.

Question 53 [+1 Mark, -1 Negative Mark]

M is the son of N. N is the sister of O. O is the father of P. How is M related to P?

A Brother

B Cousin ✓ Correct Answer

C Nephew

D Uncle

Step-by-Step Solution: N and O are brother and sister. M is N's son and P is O's child. Children of siblings are first cousins.

Question 54 [+1 Mark, -1 Negative Mark]

Statements: All the windows are doors. No door is a wall.
Conclusions:
(I) Some windows are walls.
(II) No wall is a door.

A Only I follows
B Only II follows ✓ Correct Answer
C None follows
D Both follow

Step-by-Step Solution: All windows are doors and no door is a wall => No window is a wall, so (I) is false. Since 'No door is a wall' converts directly to 'No wall is a door', conclusion (II) definitely follows.

Question 55 [+1 Mark, -1 Negative Mark]

Statements: All cups are books. All books are shirts.
Conclusions:
(I) Some cups are not shirts.
(II) Some shirts are cups.

A Only I follows
B Only II follows ✓ Correct Answer
C None follows
D Both follow

Step-by-Step Solution: All cups are books + All books are shirts => All cups are shirts. Therefore, 'Some shirts are cups' follows by conversion, and 'Some cups are not shirts' is false.

Question 56 [+1 Mark, -1 Negative Mark]

Statements: No vehicle is a bicycle. Some bicycles are electric. All electric things require power.
Conclusions:
(I) Some things requiring power are bicycles.
(II) No vehicle requires power.

A Only conclusion I follows ✓ Correct Answer
B Only conclusion II follows
C Both conclusions follow
D Neither conclusion follows

Step-by-Step Solution: Since some bicycles are electric and all electric things require power, some bicycles require power => 'Some things requiring power are bicycles' follows. Conclusion II is unsupported.

Question 57 [+1 Mark, -1 Negative Mark]

P, Q, R, S, T, U and V are seven friends sitting in a circle facing the center. V is second to the left of S and is the neighbor of T and U. S is not a neighbor of R or T. P is a neighbor of Q and R. Which of the following is correct?

A Q is between P and S ✓ Correct Answer
B S is between U and P
C T is to the immediate right of V
D U is to the immediate left of V

Step-by-Step Solution: Arranging clockwise: S, U, V, T, R, P, Q. In this layout, Q is placed right between P and S.

Question 58 [+1 Mark, -1 Negative Mark]

Six tasks A, B, C, D, E and F are arranged from first to sixth. A is before C, C is before E, B is immediately after D, and F is before A. Which task can be third?

A Task A

B Task B ✓ Correct Answer

C Task D

D Task F

Step-by-Step Solution: Order constraint: F < A < C < E and D followed immediately by B (DB). If we place F first, D second, and B third, we get F, D, B, A, C, E, which satisfies all conditions: F < A < C < E and DB contiguous. Thus, B can be third.

Question 59 [+1 Mark, -1 Negative Mark]

If DELHI is represented by its alphabetical positions as 4-5-12-8-9, how will CHENNAI be represented?

A 3-8-5-14-14-1-9 ✓ Correct Answer

B 3-8-5-13-13-1-9

C 3-7-5-14-14-1-9

D 2-8-5-14-14-1-9

Step-by-Step Solution: C = 3, H = 8, E = 5, N = 14, N = 14, A = 1, I = 9 => 3-8-5-14-14-1-9.

Question 60 [+1 Mark, -1 Negative Mark]

The HCF of two numbers is 18 and their LCM is 1260. If one number is 90, what is the other number?

A 216

B 252 ✓ Correct Answer

C 270

D 280

Step-by-Step Solution: Product of two numbers = HCF * LCM => 90 * Other = 18 * 1260 => Other = (18 * 1260) / 90 = 252.